An Axial $\ell = 2$ Non-Split Regge–Wheeler Self-Extension and a Defective Schwarzschild Resonance in Pure Weyl Gravity

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Abstract

We show that an axial $\ell = 2$ Schwarzschild quasinormal mode becomes a defective resonance in four-dimensional pure Weyl gravity. The reduced six-state Bach system has two spin-two Regge–Wheeler layers and one spin-one Maxwell layer. The repeated spin-two block is not a rational direct sum: it is a non-split self-extension. At a validated simple Regge–Wheeler quasinormal frequency, its connection has Smith type $(0, 0, 2)$, hence one length-two root chain. The geometric root is Einstein, while the generalized root has nonzero gauge-invariant traceless-Ricci carrier.

The algebraic mechanism is an exact projective cocycle. After Liouville reduction,

$$[\mathcal{I}_{\mathrm{Bach}}] = \frac{i\omega}{2}\left[1 - \frac{2}{r}\right].$$

The class in brackets is the scalar spin-two graded projection of the first signed mass-squared jet of Einstein–Weyl gravity. Its rational primitive has the same linear asymptotics as the derivative of the moving massive phase; the Coulomb exponent has zero first mass derivative. We do not identify this scalar tangent with the complete coupled massive axial system without an additional intertwining certificate.

The associated finite-interval radial Green operator, and equivalently its compactly cut-off exterior realization, has a nonzero rank-one second-order pole. Exact axial reconstruction shows that the Einstein range of the principal coefficient has nonzero Bondi shear. A complexified conserved, traceless, radially compact odd source has nonzero adjoint overlap. Thus the isolated local resonance contour contains the expected $te^{i\omega_n t}$ Einstein profile. We do not claim a global retarded quasinormal expansion, a real causal source construction, or standard asymptotic falloff for the time-independent generalized component.

1 Introduction

Fourth-order linear equations are often described by listing their scalar factors. That description is incomplete when a factor is repeated. The layers may form a direct sum, or they may form a non-semisimple extension. Only the second possibility carries an intrinsic generalized state and can turn a simple resonance of the repeated factor into a defective resonance of the full system. We use the standard Regge–Wheeler and Schwarzschild QNM conventions of Refs. [1, 2], translated to our $e^{+i\omega t}$ Fourier sign.

This distinction is realized by axial $\ell = 2$ Schwarzschild perturbations of pure Weyl gravity. In the factor-adapted radial variables, the Bach equation contains two spin-two Regge–Wheeler layers

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and one spin-one Maxwell layer, but the two spin-two layers cannot be separated by a rational local field redefinition. The extension is also the critical limit of the massless and massive spin-two directions at the level of its graded scalar tangent. At a Schwarzschild quasinormal frequency, this confluence produces a length-two root chain rather than two independent Einstein roots.

We use $M = 1$, the Fourier convention $e^{+i\omega t}$, and the negative-real-frequency representative of the reflected Schwarzschild pair. Thus damping means $\operatorname{Im}\omega > 0$, and the familiar fundamental axial frequency is written near $-0.37367 + 0.08896i$. The parameter m below is a signed *squared-mass coefficient*; it is not the positive square root usually called the graviton mass.

The paper has five results.

1. The axial spin-two Bach block is a non-split rational Regge–Wheeler self-extension and an exact first parameter jet.

2. Its projective cocycle has the elementary normal form

$$[\mathcal{I}_{\text{Bach}}] = \frac{i\omega}{2}\left[1 - \frac{2}{r}\right],$$

   equal to the scalar spin-two graded projection of the formal critical mass-squared jet.

3. A validated QNM calculation selects the defective Smith type $(0, 0, 2)$. The generalized root necessarily crosses the Einstein filtration.

4. The compact-source, compact-observation exterior outgoing radial Green operator has a genuine rank-one pole of order two.

5. Exact metric reconstruction and an explicit conserved, traceless odd source show that the range and source covectors of this pole can both be nonzero.

These statements combine exact algebra with validated numerics. The module reduction, cocycle identities, mass comparison, Green reduction, metric reconstruction, and source construction are exact. The existence and uniqueness of the enclosed QNM, nonvanishing of the projective selector, and spin-one local-unit condition are interval-validated.

Claim boundary

Result	Input	Status
Filtration, partial jet, and nonsplitting	Exact identities and exhaustive rational splitting test	`LOCAL-ALGEBRAIC`
Graded mass-direction comparison	Exact cocycle primitive and formal moving massive phases	`LOCAL-ALGEBRAIC`
Defective QNM and non-Einstein root	Validated Evans winding, selector, spin-one unit	`REDUCED-MODE`
Exterior cut-off Green double pole	Analytic Jost maps and exact transparent cuts	`REDUCED-MODE`
Bondi-shear and compact-source nonannihilation	Exact reconstruction and conserved-source identities	`REDUCED-MODE`
Global retarded ringdown expansion	Weighted Fredholm domains and contour deformation	Not established
Full coupled massive-system crosswalk	Parameter-dependent axial intertwiner and matched endpoint maps	Not established

The last row is essential. The paper proves a meromorphic exterior radial response after compact source and observation localization, and it proves the polynomial contribution of an isolated local resonance contour. It does not prove a complete causal quasinormal expansion, late-time asymptotics, detector sensitivity, or excitation by a specified astrophysical matter model.

The nearby quadratic-gravity problem has been studied directly as a coupled massive spin-two system [3, 4]. That literature separates the general-relativistic and massive QNM families at nonzero squared mass. Our distinct result concerns their singular pure-Weyl confluence and proves non-semisimplicity of the critical Bach response. We do *not* replace the complete massive axial system by a scalar equation; the scalar mass equation used below is only its spin-two graded tangent target.

2 The filtered axial Bach module

2.1 Differential-module conventions

We work over

$$K = \mathbb{C}(r, \omega), \qquad D = f\partial_r, \qquad f = 1 - \frac{2}{r},$$

with $\omega \neq 0$ unless stated otherwise. The spin-two and spin-one scalar operators for axial $\ell = 2$ are

$$L_2 = D^2 + U_2, \qquad U_2 = \omega^2 - f\left(\frac{6}{r^2} - \frac{6}{r^3}\right),$$

$$L_1 = D^2 + U_1, \qquad U_1 = \omega^2 - f\frac{6}{r^2}.$$

We use $D(q)$ and $D(U)$, rather than primes, for the radial derivation. A prime on a connection coefficient below denotes ∂_ω.

The Einstein submodule is the kernel of the gauge-invariant traceless-Ricci, equivalently Schouten, carrier. Thus "non-Einstein" means nonzero projection to this carrier quotient, not failure of a coordinate gauge choice.

2.2 Filtration and partial jet

In the certified factor gauge, the six-state system is

$$\begin{aligned} DX &= AX + EY + CZ, \\ DY &= AY + DZ, \\ DZ &= A_1 Z, \end{aligned} \tag{1}$$

where $X, Y, Z \in K^2$, while A and A_1 are the companion matrices of L_2 and L_1. The rational reconstruction into the axial metric and curvature variables has full row rank on the declared domain.

Theorem 2.1 (Filtered axial Bach module)**.** *The reduced axial $\ell = 2$ Bach module has a filtration with successive quotients*

$$M_2, \qquad M_2, \qquad M_1.$$

The middle four-state object is an extension

$$0 \longrightarrow M_2 \longrightarrow E_2 \longrightarrow M_2 \longrightarrow 0,$$

and the final quotient is the spin-one Maxwell module M_1.

This theorem is an exact rational statement. It does not rely on an endpoint normalization or on the QNM calculation.

Introduce the four-state family

$$\mathcal{B}(\tau) = \begin{pmatrix} A+\tau E & D+\tau C \\ 0 & A_1 \end{pmatrix}, \qquad \Phi_4(\tau) = \begin{pmatrix} P(\tau) & Q(\tau) \\ 0 & R \end{pmatrix}.$$

Theorem 2.2 (Exact partial jet)**.** *The system (1) is the first spin-two-row jet of $\mathcal{B}(\tau)$ at $\tau = 0$. Its fundamental matrix is*

$$\Phi_6 = \begin{pmatrix} P & \dot{P} & \dot{Q} \\ 0 & P & Q \\ 0 & 0 & R \end{pmatrix}_{\tau=0}, \qquad \det \Phi_6 = (\det P)^2 \det R.$$

Every factor-adapted endpoint connection consequently has the form

$$T(\omega) = \begin{pmatrix} a & b & c \\ 0 & a & d \\ 0 & 0 & g \end{pmatrix}, \qquad \det T = a^2 g. \tag{2}$$

Although each scalar factor is second order, its chosen horizon or infinity condition selects a one-dimensional endpoint line. The matrix (2) records one unwanted scalar endpoint coefficient for each of the three admissible factor lines; it is therefore 3×3, rather than the 6×6 fundamental matrix of the first-order radial system.

Proof. Differentiate $D\Phi_4(\tau) = \mathcal{B}(\tau)\Phi_4(\tau)$ at $\tau = 0$. The differentiated first row, the undifferentiated spin-two row, and the spin-one row reproduce (1). The determinant and connection identities follow from block triangularity. □

2.3 Projective extension class

Consider a scalar self-extension

$$Lx = (s_1 D + s_0)y, \qquad Ly = 0, \qquad L = D^2 + U.$$

Define

$$\mathcal{I} = s_0 - \frac{1}{2}D(s_1), \qquad \mathcal{K}_U = D^3 + 4UD + 2D(U).$$

Lemma 2.3 (Triangular gauge)**.** *For $q \in K$, set*

$$Q(q) = qD - \frac{1}{2}D(q).$$

On $\ker L$*,*

$$[L, Q(q)] = -\frac{1}{2}\mathcal{K}_U(q).$$

Thus the lift change $x \mapsto x + Q(q)y$ changes the scalar density by

$$\mathcal{I} \longmapsto \mathcal{I} - \frac{1}{2}\mathcal{K}_U(q).$$

Proof. Expand the commutator by the Leibniz rule and use $D^2 y = -Uy$. The coefficient of Dy cancels; the remaining coefficient is $-[D^3 q + 4UDq + 2D(U)q]/2$. □

The projective extension class is therefore

$$[\mathcal{I}] \in K/\mathcal{K}_U K.$$

The operator $\mathcal{K}_U$ is the symmetric-square operator of L.

Theorem 2.4 (Mass-direction normal form)**.** *For the axial $\ell = 2$ Bach extension,*

$$\mathcal{I}_{\mathrm{Bach}} = \frac{i(r-2)(2r\omega^2 + 3\omega^2 + 12)}{5r^4\omega}.$$

Let

$$q(r,\omega) = -\frac{i}{120\omega}\left(15r + 13 + \frac{12}{r} + \frac{9}{r^2}\right).$$

Then, for $\omega \neq 0$,

$$\mathcal{I}_{\mathrm{Bach}} = \mathcal{K}_{U_2} q + \frac{i\omega}{2} f, \tag{3}$$

and hence

$$\boxed{[\mathcal{I}_{\mathrm{Bach}}] = \frac{i\omega}{2}[f].}$$

Proof. Substitute $D = f\partial_r$, U_2, and the displayed q into $\mathcal{K}_{U_2} q$. After clearing $120r^4\omega$, the assertion is a polynomial identity in r and ω. Exact coefficient comparison gives (3). □

Theorem 2.5 (Non-split physical-axis extension)**.** *For every real $\omega > 0$, the extension E_2 is non-split over $\mathbb{C}(r)[D]$.*

Certificate-supported exact proof. The local pole audit permits poles only at $r = 0$ and at the apparent divisor $r\omega = 2i$. Its indicial bounds force the exhaustive ansatz

$$a(r) = -\frac{i}{\omega^2(r\omega - 2i)} + \frac{c_{-2}}{r^2} + \frac{c_{-1}}{r} + c_0.$$

After substitution in $\mathcal{K}_{U_2} a + 2\mathcal{I}_{\mathrm{Bach}} = 0$ and clearing the declared denominators, coefficient comparison gives

$$M(\omega)c = b(\omega), \qquad M \in \mathbb{C}^{6\times 3},$$

with

$$M = \begin{pmatrix} 0 & -4\omega^4 & 0 \\ -8\omega^4 & 8\omega^4 & 24\omega^2 \\ 16\omega^4 & 42\omega^2 & -156\omega^2 \\ 48\omega^2 & -224\omega^2 & 312\omega^2 \\ -208\omega^2 & 364\omega^2 & -192\omega^2 \\ 224\omega^2 & -168\omega^2 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ -2i(5\omega - 6i) \\ i(35\omega - 78i) \\ -6i(5\omega - 26i) \\ 96 \\ 0 \end{pmatrix}.$$

Fixed minors of the coefficient and augmented matrices are

$$\det M_{[0,1,2]} = 3456\,\omega^{10}, \qquad \det[M \mid \mathrm{rhs}]_{[0,1,2,5]} = -645120\,i\,\omega^9.$$

They are simultaneously nonzero for every real $\omega > 0$. Hence $\operatorname{rank} M = 3$, whereas $\operatorname{rank}[M \mid b] \geq 4$, so the inhomogeneous coboundary equation is inconsistent. The pole audit and displayed ansatz are what make this finite inconsistency exhaustive rather than a truncated search. □

Theorem 2.5 is a statement about the differential module; Theorem 2.4 identifies a simple representative of its class. Neither theorem asserts that every QNM of the repeated scalar factor must be defective. Defectiveness is a resonant evaluation of the nonzero extension class.

3 Graded mass-squared tangent and crosswalk boundary

3.1 The scalar graded target

The covariant parent family contains the second-order tensor pencil

$$(\mathcal{E}+m\mathcal{A})\phi=0,$$

where m is the signed squared-mass coefficient. On the transverse-traceless spin-two graded quotient, the scalar target is

$$L^{\mathrm{gr}}_m(\omega)y=\left[D^2+\omega^2-V_2-mf\right]y=0. \tag{4}$$

This is *not* asserted to be the complete coupled massive axial system of Refs. [3, 4]. It is the spin-two graded projection used to compare first parameter jets.

Proposition 3.1 (Exact graded mass direction)**.** *Differentiating* (4) *at* $m=0$ *gives*

$$L_2\,\partial_m y=f\,y,\qquad [\mathcal{I}^{\mathrm{gr}}_{\mathrm{mass}}]=[f].$$

Consequently,

$$\boxed{[\mathcal{I}_{\mathrm{Bach}}]=\frac{i\omega}{2}[\mathcal{I}^{\mathrm{gr}}_{\mathrm{mass}}].}$$

Equivalently, at each fixed $\omega\neq 0$*, the two tangent cocycles are related by* $m=(i\omega/2)\tau$*, modulo rational triangular gauge. Because this factor depends on the spectral variable, it is not a global reparametrization of two operator families.*

Proof. Differentiating (4) gives the extension density f. The result follows from Theorem 2.4. □

3.2 Formal endpoint comparison

Because Lemma 2.3 contains a factor 1/2, the field redefinition that removes $\mathcal{K}_{U_2}q$ is

$$Q_q=2qD-D(q),\qquad [L_2,Q_q]|_{\ker L_2}=-\mathcal{K}_{U_2}(q).$$

Thus $L_2x=\mathcal{I}_{\mathrm{Bach}}y$ implies

$$L_2(x+Q_qy)=\frac{i\omega}{2}fy.$$

The primitive has the forced behavior

$$q(r,\omega)=-\frac{i}{8\omega}r+O(1).$$

Proposition 3.2 (Formal differentiated endpoint asymptotics)**.** *Choose the square-root branch*

$$k=(\omega^2-m)^{1/2},\qquad k(\omega,0)=\omega,$$

and the formal infinity branches

$$y^{\mathrm{gr}}_\sigma(m,r)=e^{\sigma ikr}r^{\rho_\sigma(m)}\left(1+O(r^{-1})\right),\qquad \rho_\sigma(m)=\sigma i\left(2k+\frac{m}{k}\right).$$

Then

$$\rho'_\sigma(0)=0,\qquad \frac{\partial_m y^{\mathrm{gr}}_\sigma(0,r)}{y^{\mathrm{gr}}_\sigma(0,r)}=-\frac{\sigma i}{2\omega}r+O(1),$$

and, with $c(\omega) = i\omega/2$,

$$\frac{Q_q y_\sigma}{y_\sigma} = \frac{\sigma r}{4} + O(1) = c(\omega)\frac{\partial_m y_\sigma^{\mathrm{gr}}(0,r)}{y_\sigma^{\mathrm{gr}}(0,r)} + O(1).$$

At the horizon, $f = O(r-2)$, so the graded mass term does not change the ingoing Frobenius exponent.

Proof. The infinity equation has

$$D^2 y + \left(k^2 + \frac{2m}{r} + O(r^{-2})\right) y = 0.$$

The coefficient of r^{-1} gives

$$2\sigma i k \rho_\sigma + 4k^2 + 2m = 0.$$

Since $k'(0) = -1/(2\omega)$,

$$\frac{d}{dm}\left(2k + \frac{m}{k}\right)\bigg|_{m=0} = 0.$$

The Coulomb logarithm therefore cancels at first order, while the moving plane-wave phase supplies the linear term. The formula for Q_q follows from $Dy_\sigma/y_\sigma = \sigma i\omega + O(r^{-1})$. $\square$

The proposition is deliberately formal. It verifies the endpoint asymptotic target but does not exclude an opposite-Jost admixture in an exact massive solution. Promotion to a physical mass-velocity theorem requires (i) a displayed parameter-dependent map from the complete coupled massive axial variables to the certified filtration, and (ii) a Volterra or equivalent uniqueness construction for the mass-differentiated horizon and infinity maps. Neither object is imported as a certificate here.

Remark 3.3 (Conditional physical crosswalk). If those two objects are supplied, scalar endpoint normalizations can only add an analytic multiple of the unperturbed Evans coefficient. One would then obtain

$$b_{\mathrm{B}}(\omega) = \frac{i\omega}{2}\partial_m a(\omega,0) + h(\omega)a(\omega,0),$$

and, at a simple QNM,

$$\omega_n'(0) = \frac{2i}{\omega_n}\kappa_n, \qquad \kappa_n = \frac{b_{\mathrm{B}}(\omega_n)}{a_\omega(\omega_n,0)}.$$

These formulas are the precise remaining crosswalk target, not a theorem of the present paper.

The graded construction is the asymptotically flat analogue of the mass-derivative mechanism behind logarithmic partners in critical gravity [5, 6]. Here the formal endpoint tangent is polynomial rather than an independent radial $\log r$ mode.

4 The defective Schwarzschild resonance

4.1 Smith dichotomy

At a simple spin-two zero, analytic elimination of the spin-one unit reduces (2) to

$$T_2(\omega) = \begin{pmatrix} a(\omega) & b(\omega) \\ 0 & a(\omega) \end{pmatrix}, \qquad a(\omega_n) = 0, \quad a'(\omega_n) \neq 0.$$

All Smith forms in this section are taken over the discrete valuation ring

$$\mathcal{O}_{\omega_n} = \{\text{holomorphic germs at } \omega_n\},$$

whose uniformizer is $z = \omega - \omega_n$. The listed integers are the valuations of the invariant factors. Their sum is the algebraic multiplicity; the number of positive invariant factors is the geometric root-space dimension only after the unit factors have been removed. This is the local analytic-matrix setting underlying Keldysh root chains and nonlinear eigenvalue pencils [8, 9].

Proposition 4.1 (Local Smith dichotomy)**.** *If $g(\omega_n) \neq 0$, then*

$$\mathrm{Smith}(T) = \begin{cases} (0,0,2), & b(\omega_n) \neq 0, \\ (0,1,1), & b(\omega_n) = 0. \end{cases}$$

The first case has one length-two root chain; the second has two independent length-one roots.

Proof. With $z = \omega - \omega_n$, $a = a_1 z + O(z^2)$, $a_1 \neq 0$. The determinant valuation of the spin-two block is two. If $b(\omega_n) \neq 0$, its first Fitting ideal is the unit ideal and the exponents are $(0, 2)$. If $b(\omega_n) = 0$, all entries vanish to first order and the exponents are $(1, 1)$. The spin-one unit contributes a zero exponent. □

4.2 Validated specialization

Transport the horizon-ingoing and infinity-outgoing spin-two lines to the same ordinary matching section. In the Riccati chart $v = Y_2/Y_1$, define

$$\delta(\omega, \tau) = v_\infty(\omega, \tau) - v_H(\omega, \tau), \qquad \delta_\tau = \partial_\tau \delta|_{\tau=0}\,.$$

The scalar Evans coefficient is $a = u\delta$ for a holomorphic unit u. At a simple zero,

$$\frac{b(\omega_n)}{a'(\omega_n)} = \frac{\delta_\tau(\omega_n)}{\delta_\omega(\omega_n)}.$$

This definition removes arbitrary homogeneous amplitudes; a projective chart change multiplies numerator and denominator by the same nonzero value at the root.

The certified complex-scaled Volterra construction makes both selected lines holomorphic throughout the closed disk bounded by Γ. The disk lies inside the declared continuation strip and avoids every listed Frobenius and normalization pole. Endpoint series are initialized with outward remainder bounds; a failed remainder, chart transition, or panel overlap is a rejection. Consequently δ is holomorphic, not meromorphic, on the argument-principle domain.

The counterclockwise contour

$$\Gamma(t) = \omega_c + \frac{1}{40} e^{2\pi i t}, \qquad 0 \leq t \leq 1,$$

is split into 927 validated panels. Every panel excludes zero, the overlapping argument sectors have width below π, and their unique continuous lift changes by 2π. A local Rouché calculation then encloses the unique root in

$$\Omega_n = \left\{\omega : \left|\omega - \left(-0.3736716844180418 + 0.0889623156889357\, i\right)\right| \leq 10^{-7}\right\}.$$

Projective interval panels exclude zeros on the contour and give winding number $+1$. The disk therefore contains exactly one simple spin-two zero. On the localized disk,

$$\delta_\tau(\Omega_n) \subset [0.0010932 \pm 7.60 \times 10^{-8}] + i[-0.0002195 \pm 8.89 \times 10^{-8}],$$

which excludes zero. An independent local calculation bounds the spin-one mismatch modulus below by 1/2500.

The local transport uses outward-rounded `python-flint acb/arb` balls at 60 decimal digits, order-48 Taylor transport, 966 accepted horizon steps, and 260 accepted outgoing steps. The frequency derivative obeys

$$\delta_\omega(\Omega_n) \subset [\pm 4.88 \times 10^{-3}] + i[-0.0100 \pm 1.82 \times 10^{-3}],$$

and therefore excludes zero. Direct interval division gives the dimensionless intrinsic selector

$$\boxed{\operatorname{Re}\kappa_n \in [-0.047, 0.022], \qquad \operatorname{Im}\kappa_n \in [0.064, 0.138].} \tag{5}$$

The enclosure is deliberately reported as an inequality rather than a long decimal approximation. This is an interval-validated Evans/argument-principle computation, not a floating-point winding inferred from sampled phases; compare the general Evans-function methodology in Ref. [10].

Theorem 4.2 (Certified defective resonance)**.** *At the unique QNM in Ω_n, the complete axial connection has Smith valuations*

$$\boxed{(0, 0, 2).}$$

Its spin-two divisor has algebraic multiplicity two, geometric multiplicity one, and one length-two root chain.

Proof. The winding certificate supplies the simple spin-two zero. The selector enclosure gives $b(\omega_n) \neq 0$, and the spin-one bound gives $g(\omega_n) \neq 0$. Proposition 4.1 applies. □

4.3 One resonant invariant

Encode the endpoint conditions in an analytic finite-interval pencil

$$L(\omega) : X \longrightarrow Y.$$

Such holomorphic boundary pencils and their contour/root-chain reductions fit the analytic Fredholm framework of Refs. [11, 12]. Choose right and left states at the simple scalar resonance,

$$L_n u_n = 0, \qquad L_n^* \widetilde{u}_n = 0,$$

and define, with all augmented endpoint contributions included,

$$\alpha_n = \langle \widetilde{u}_n, L_n' u_n \rangle \neq 0.$$

For the repeated block

$$\mathbb{L}(\omega) = \begin{pmatrix} L(\omega) & K(\omega) \\ 0 & L(\omega) \end{pmatrix},$$

set

$$\beta_n = \langle \widetilde{u}_n, K_n u_n \rangle.$$

Theorem 4.3 (Resonant evaluation)**.** *In endpoint frames compatible with the partial jet,*

$$\boxed{\kappa_n = \frac{b(\omega_n)}{a'(\omega_n)} = \frac{\beta_n}{\alpha_n} = -\left.\frac{d\omega_n}{d\tau}\right|_0 .} \tag{6}$$

The ratio is invariant under a common analytic nonzero Evans unit and under triangular changes of splitting.

Proof. Differentiate

$$[L(\omega) + \tau K(\omega)]u(\tau) = 0$$

along the resonance branch. Pairing with $\widetilde{u}_n$ gives

$$\dot{\omega}_{n,\tau}\alpha_n + \beta_n = 0, \qquad \dot{\omega}_{n,\tau} := \left.\frac{d\omega_n}{d\tau}\right|_0 .$$

Differentiating the connection divisor gives

$$a'(\omega_n)\dot{\omega}_{n,\tau} + b(\omega_n) = 0,$$

because the partial jet identifies $b = \partial_\tau a$. The mass crosswalk is not used. A triangular change $K \mapsto K + LQ - QL$ has zero resonant pairing, and an Evans unit multiplies numerator and denominator by the same nonzero value. □

Under the additional hypotheses of Remark 3.3, the last member of (6) may also be written $-i\omega_n\omega_n'(0)/2$, where the derivative is with respect to the physical signed squared-mass coefficient. The certified enclosure (5) is intrinsic; it is not advertised here as a massive-QNM slope.

Proposition 4.4 (Non-Einstein generalized root)**.** *The geometric root belongs to the Einstein submodule. Every length-two root polynomial has generalized coefficient with nonzero projection to the Ricci-carrier quotient. In the normalized triangular frame, that projection is*

$$\boxed{-\frac{a'(\omega_n)}{b(\omega_n)} = -\frac{\alpha_n}{\beta_n} = -\frac{1}{\kappa_n} \neq 0.}$$

Proof. Write $a = a_1 z + O(z^2)$, $b = b_0 + O(z)$, with $a_1 b_0 \neq 0$. For a root polynomial

$$c(z) = c_0 + zc_1 + O(z^2), \qquad T_2(z)c(z) = O(z^2),$$

the order-zero equation gives $c_0 = e_X$, the Einstein vector. Writing $c_1 = xe_X + ye_Y$, the order-z equation is

$$a_1 + b_0 y = 0.$$

Hence $y = -a_1/b_0 \neq 0$. Adding an Einstein vector changes x, not the carrier coefficient y. □

Thus the defective resonance is not the same Einstein QNM counted twice. Its generalized member is the canonical mass-tangent class modulo the ordinary QNM.

5 Exterior Green pole and critical response

5.1 Finite cuts and the exterior inverse

Put $x = r_*$. On the certified QNM neighborhood, let

$$u_H(x, \omega), \qquad u_\infty(x, \omega)$$

be analytic spin-two Jost solutions satisfying the future-horizon and outgoing-infinity conditions. Their Wronskian is the scalar Evans coefficient, up to an analytic unit. Our cut-off exterior terminology is consistent with rigorous asymptotically flat resonance definitions through meromorphically continued cut-off resolvents [13, 14]; the present theorem is only the local radial realization described below.

Choose nontrivial smooth source and observation cutoffs $\chi_{\mathrm{s}}, \chi_{\mathrm{o}}$ and finite cuts

$$x_- < \operatorname{supp}\chi_{\mathrm{s}} \cup \operatorname{supp}\chi_{\mathrm{o}} < x_+.$$

Impose the exact transparent boundary conditions

$$W(u, u_H)|_{x_-} = 0, \qquad W(u, u_\infty)|_{x_+} = 0.$$

They are analytic in ω, and uniqueness of the exterior Jost problems identifies the finite-interval inverse between the cutoffs with $\chi_{\mathrm{o}} R_{\mathrm{ext}}(\omega)\chi_{\mathrm{s}}$.

Theorem 5.1 (Exterior cut-off Green pole)**.** *For any such nontrivial compact source and observation cutoffs,*

$$\chi_{\mathrm{o}} R_{\mathrm{ext}}(\omega)\chi_{\mathrm{s}}$$

has a nonzero rank-one pole of order two at ω_n. In the augmented normalization,

$$\boxed{\operatorname{Coeff}_{(\omega-\omega_n)^{-2}}\big(\chi_{\mathrm{o}} R_{\mathrm{ext}}\chi_{\mathrm{s}}\big) = -\frac{\beta_n}{\alpha_n^2}(\chi_{\mathrm{o}} u_n) \otimes (\widetilde{u}_n \chi_{\mathrm{s}}).} \tag{7}$$

Proof. The repeated block inverse is

$$\mathbb{L}^{-1} = \begin{pmatrix} L^{-1} & -L^{-1}KL^{-1} \\ 0 & L^{-1} \end{pmatrix}.$$

At a simple scalar resonance,

$$L(\omega)^{-1} = \frac{u_n \otimes \widetilde{u}_n}{\alpha_n(\omega - \omega_n)} + O(1).$$

The upper-right block therefore has principal coefficient

$$-\frac{\beta_n}{\alpha_n^2} u_n \otimes \widetilde{u}_n.$$

The exact transparent conditions identify this finite-interval coefficient with the cut-off exterior coefficient. It is nonzero because $\beta_n \neq 0$, and a nonzero solution of a second-order ODE cannot vanish on a nonempty open interval. □

The full doubled-state principal coefficient maps the carrier channel to the Einstein channel and vanishes on the Einstein channel; it is square-zero on that graded space. Its projection to a single metric or radial block is rank one but is not intrinsically a nilpotent operator after the grading has been forgotten.

5.2 Parent normalization and isolated contour

Our normalization of α_{W} is fixed by the quadratic parent action

$$S_m^{(2)} = 4\alpha_{\mathrm{W}} \int \left[\phi \cdot \mathcal{E}(h) - \frac{1}{2}\phi \cdot \mathcal{A}(\phi) + \frac{m}{2} h \cdot \mathcal{E}(h)\right].$$

The equations are

$$\mathcal{E}(h) = \mathcal{A}(\phi), \qquad \mathcal{E}(\phi) + m\mathcal{E}(h) = 0.$$

Eliminating the auxiliary field and inverting the $m = 0$ block gives

$$G^{\mathrm{W}}_{hh} = \frac{1}{4\alpha_{\mathrm{W}}}\mathcal{E}^{-1}\mathcal{A}\mathcal{E}^{-1}.$$

At points where the second-order tensor inverses exist,

$$\partial_m(\mathcal{E} + m\mathcal{A})^{-1} = -(\mathcal{E} + m\mathcal{A})^{-1}\mathcal{A}(\mathcal{E} + m\mathcal{A})^{-1}.$$

Consequently

$$\boxed{G^{\mathrm{W}}_{hh}(\omega) = -\frac{1}{4\alpha_{\mathrm{W}}}\,\partial_m(\mathcal{E} + m\mathcal{A})^{-1}\big|_{m=0}\,.} \tag{8}$$

No commutativity assumption is used.

If a fixed-domain massive tensor pencil has the local simple-pole form

$$R_m(\omega) = \frac{P_n(m)}{\omega - \omega_n(m)} + H_n(\omega, m),$$

then (8) gives

$$G^{\mathrm{W}}_{hh}(\omega) = -\frac{1}{4\alpha_{\mathrm{W}}}\left[\frac{\omega_n'(0)P_n(0)}{(\omega - \omega_n)^2} + \frac{P_n'(0)}{\omega - \omega_n} + \partial_m H_n(\omega, 0)\right].$$

Thus the double pole measures QNM motion; the accompanying simple pole measures motion of the residue projector. This exact parent identity does not, by itself, identify the certified scalar κ_n with the physical massive-QNM velocity; that is the crosswalk left open in Remark 3.3.

Corollary 5.2 (Isolated resonance contour)**.** *Write the certified radial inverse as*

$$R_{\mathrm{ext}}(\omega) = \frac{\Pi_{-2}}{(\omega - \omega_n)^2} + \frac{\Pi_{-1}}{\omega - \omega_n} + O(1)$$

between compact source and observation cutoffs. For a contour enclosing only this local resonance,

$$\frac{1}{2\pi i}\oint e^{i\omega t} R_{\mathrm{ext}}(\omega)\, d\omega = e^{i\omega_n t}\left(\Pi_{-1} + it\,\Pi_{-2}\right).$$

The t-weighted spatial profile is the ordinary Einstein QNM, while the constant generalized component crosses the carrier quotient.

This is a local residue theorem, not a global retarded expansion. The latter would require a Laplace contour, a closed weighted realization, control of thresholds and non-pole contributions, and a justified contour deformation.

6 Source and observation channels

6.1 Metric reconstruction at null infinity

Let E be the unit outgoing Einstein line and R the unit outgoing carrier line in the exact factor frame. After factoring the common retarded phase $e^{i\omega u}$, their Regge–Wheeler-gauge metric heads are

$$(H_0, H_1)_E = (-r, 2r) + O(1),$$
$$(H_0, H_1)_R = \left(\frac{3}{4}r^2 - \frac{3}{2}r, -\frac{3}{2}r^2\right) + O(1).$$

Hence

$$(H_0, H_1)_R = -\frac{3}{4} r\,(H_0, H_1)_E + O(E).$$

In odd radiation gauge,

$$h_u \mapsto h_u - i\omega\xi, \qquad h_2 \mapsto h_2 - 2\xi, \qquad \xi = \frac{h_u}{i\omega},$$

so $h_2 = 2ih_u/\omega$. Writing $h_{AB} = h_2 X_{AB}$, define the Bondi-shear observation map by

$$\mathcal{O}_{\mathscr{I}^+} h = \lim_{r\to\infty} \frac{h_{AB}}{r}.$$

Theorem 6.1 (Null-infinity reconstruction)**.** *The outgoing Einstein line has nonzero Bondi shear,*

$$\boxed{\mathcal{O}_{\mathscr{I}^+} E = -\frac{2i}{\omega} X_{AB} \neq 0.} \tag{9}$$

Its strain has the standard radiative falloff

$$\frac{h^E_{AB}}{r^2} = -\frac{2i}{\omega r} e^{i\omega u} X_{AB} + O(r^{-2}).$$

The carrier line instead has

$$\boxed{\frac{h^R_{AB}}{r^2} = \frac{3i}{2\omega} e^{i\omega u} X_{AB} + O(r^{-1}).} \tag{10}$$

Therefore the time-independent generalized component has weakened $O(1)$ strain in this standard odd radiation gauge. The Einstein range of the double-pole coefficient is nevertheless observed nontrivially at future null infinity.

Proof. Substitute the exact outgoing factor-frame series into the algebraic metric reconstruction. The odd gauge law then gives (9) and (10). Adding an Einstein representative cannot cancel the carrier r^2 coefficient, which is one power higher. The trace-free odd tensor harmonic cannot be removed by a Weyl rescaling; its removal would require a large boundary diffeomorphism rather than a standard small gauge transformation. □

Proposition 6.2 (Outgoing-trace bridge)**.** *On the local QNM disk, coefficient extraction from the selected outgoing Jost family defines a holomorphic trace*

$$\mathcal{T}_{\mathscr{I}^+} : u \longmapsto \textit{outgoing asymptotic coefficient.}$$

The algebraic reconstruction followed by $\mathcal{T}_{\mathscr{I}^+}$ is therefore an analytic observation map. For compactly supported forcing,

$$\mathrm{Coeff}_{(\omega-\omega_n)^{-2}}\big(\mathcal{O}_{\mathscr{I}^+} R_{\mathrm{ext}} \chi_{\mathrm{s}}\big) = -\frac{\beta_n}{\alpha_n^2} \mathcal{O}_{\mathscr{I}^+}(u_n) \otimes (\widetilde{u}_n \chi_{\mathrm{s}}).$$

In particular, Theorem 6.1 implies that the first factor is nonzero.

Proof. The outgoing coefficients and reconstruction matrices are holomorphic on the local disk. Applying a holomorphic map to a Laurent expansion acts coefficientwise on its principal part. Substitute (7) and then (9). □

The gauge parameter used above is allowed only as an algebraic reconstruction into the chosen odd radiation gauge. For the Einstein line it preserves the standard Bondi falloff. For the carrier line it is large relative to the usual asymptotically flat phase space, and (10) is consequently not claimed to define finite Bondi flux or an admissible asymptotic state.

Combining Theorem 5.1 with Proposition 6.2, the principal Bondi-shear coefficient is nonzero. If the additional physical mass crosswalk of Remark 3.3 is imposed, its parent-normalized form may be written

$$\mathcal{O}_{\mathscr{I}^+}G_{-2} = \frac{i\,\omega_n'(0)}{2\alpha_{\mathrm{W}}\alpha_n\omega_n} X_{AB} \otimes \widetilde{u}_n.$$

The isolated contour therefore contains an ordinary u/r radiative profile multiplied by the adjoint source overlap.

6.2 A complexified conserved, traceless source

Let $\ell \geq 2$, $\mu = (\ell-1)(\ell+2)$, and $\omega \neq 0$. Use the Martel–Poisson odd source projections and orientation $\epsilon_{tr} = 1$, $\epsilon^{tr} = -1$ [7].

Theorem 6.3 (Complexified conserved-source nonannihilation). *For any*

$$F \in C_c^\infty((2M,\infty)),$$

define

$$P_t = 0, \qquad P_r = \frac{\mu F}{2i\omega r f}, \qquad P = \frac{\partial_r(rF)}{2i\omega}.$$

Then

$$\nabla_a P^a + \frac{2}{r} r_a P^a - \frac{\mu}{r^2} P = 0,$$

and the Cunningham–Price–Moncrief source is

$$S_{\mathrm{odd}} = -\frac{2r}{\mu}\epsilon^{ab}\nabla_a P_b = \frac{F}{f}.$$

The associated odd stress tensor

$$T^{aB} = \frac{P^a}{16\pi r^2} X^B, \qquad T^{AB} = \frac{P}{8\pi r^4} X^{AB},$$

with all other harmonic components zero, is smooth and radially compact, and satisfies

$$\boxed{\nabla_\mu T^{\mu\nu} = 0, \qquad T^\mu{}_\mu = 0.}$$

The complexified conserved-traceless odd source map is therefore onto the smooth compact reduced radial forcings.

Moreover, choose a nonnegative nonzero $\eta \in C_c^\infty((2M,\infty))$ supported where the adjoint state $\widetilde{u}_n$ is nonzero and set

$$F = \eta\,\overline{\widetilde{u}_n}.$$

Then the augmented pairing has no endpoint term and

$$\boxed{\langle \widetilde{u}_n, F\rangle_{\mathrm{aug}} = \int \eta |\widetilde{u}_n|^2 \, dr_* > 0.}$$

Thus the space of complexified frequency-domain conserved and traceless odd sources does not annihilate the certified double pole.

Proof. Since $\sqrt{-\det g_{ab}} = 1$,

$$P^r = fP_r = \frac{\mu F}{2i\omega r}, \qquad \nabla_a P^a = \partial_r P^r.$$

Direct substitution proves the reduced conservation identity and $fS_{\mathrm{odd}} = F$. The remaining spacetime conservation equations vanish by the divergence-free odd vector harmonic, and tracelessness follows from $T^{ab} = 0$ and $\Omega_{AB} X^{AB} = 0$. Compact support removes all endpoint terms from the adjoint pairing, and the final integral is strictly positive. □

Remark 6.4 (What the source theorem does not say). The choice $F = \eta\overline{\tilde{u}_n}$ depends on the complex QNM frequency. It is an existence proof in the complexified frequency-domain source space, not a real, causal, temporally compact matter source. A real formal perturbation can be assembled by adjoining the reflected frequency and harmonic sector, but causality and temporal support still require an inverse-Laplace construction. The theorem also does not identify a specified material trajectory or an energy-condition-satisfying model. A standard massive point-particle stress tensor is not trace-free and cannot be coupled directly to the Bach equation without a conformal completion or compensator.

7 Scope, evidence, and open problem

7.1 The remaining analytic problem

The exterior cut-off theorem is already a differential-response result on the Schwarzschild exterior. The remaining causal problem is narrower but substantive. A global realization must provide:

1. closed weighted domains implementing horizon-ingoing and differentiated outgoing Jost conditions;
2. a Fredholm pencil of index zero with a fixed analytic domain;
3. bounded reconstruction through the physical gauge quotient;
4. a causal Laplace contour and a justified deformation through the QNM disk;
5. control of threshold, branch-cut, and non-pole contributions.

The weakened falloff in (10) explains why this is not routine. A standard outgoing asymptotically flat function space may contain the Einstein root but exclude its intrinsic generalized tangent. One may instead use an enlarged tangent domain or keep the bulk space fixed and encode the differentiated endpoint germ as a finite-dimensional boundary component. This paper uses the latter construction locally but does not complete its global causal implementation.

7.2 Certificate dependency

Certificate family	Imported statement	Type
Axial triangular preflight	Filtration, factor gauge, and reconstruction rank	Exact
Projective cocycle	Cocycle, primitive, and rational nonsplitting obstruction	Exact
Critical parent jet	Covariant mass derivative; physical scalar crosswalk remains open	Exact with gate
Analytic continuation	Analytic horizon and infinity Jost germs on the QNM disk	Exact gate
Full Evans contour	Zero-free contour and winding +1	Validated
Local selector	Nonzero projective tangent on the root disk	Validated
Spin-one local unit	Nonzero spin-one mismatch	Validated
Fredholm promotion	Finite-interval Poisson/trace reduction	Exact reduction
Null-infinity reconstruction	Metric heads, radiation gauge, and Bondi shear	Exact
Conserved source	Conservation, tracelessness, and adjoint overlap	Exact

The generated claim map pins the authority files and their hashes. Its verifier independently checks the displayed cocycle identity, the fixed-frequency graded tangent normalization, the Smith and Green formulas, the reconstruction falloffs, the source conservation law, and explicit non-results. Missing data, a stale hash, an unsupported domain, or a failed mutation is a rejection rather than a partial pass.

7.3 Reproduction and trust boundary

The exact certificates use Python integer/rational arithmetic and SymPy 1.14.0 for polynomial reduction. Validated complex numerics use `python-flint` 0.9.0 `arb`/`acb` balls with directed outward rounding. The reviewed environment uses Python 3.12.13 and `jsonschema` 4.26.0. Certificate parsing, these arithmetic libraries, the Python runtime, and the operating system form the trusted computing base. Displayed decimal intervals are outward enclosures; the machine records retain midpoint and radius strings that round-trip through `arb`.

The principal independent verification commands are:

Claim	Command
Projective cocycle and nonsplitting	`python3-mblack_hole_programme.phase3.axial_qnm_projective_cocycle_v1.verify`
Evans winding	`python3-mblack_hole_programme.phase3.axial_qnm_projective_evans_contour_completion.full_contour_winding_v1.verify`
Local selector	`python3-mblack_hole_programme.phase3.axial_qnm_projective_evans_contour_completion.local_selector_v1.verify`
Spin-one unit	`python3-mblack_hole_programme.phase3.axial_qnm_spin_one_local_unit_v1.verify`
Radial Green pole	`python3-mblack_hole_programme.phase4.axial_qnm_fredholm_promotion_v1.verify`
Null-infinity reconstruction	`python3-mblack_hole_programme.phase4.axial_qnm_null_infinity_reconstruction_v1.verify`
Complexified source overlap	`python3-mblack_hole_programme.phase4.axial_qnm_conserved_source_overlap_v1.verify`

The accompanying claim-map JSON records exact paths, SHA-256 hashes, producer and verifier commands, and mutation tests. A Git commit is content-addressed but is not an archival preservation service. An immutable DOI-bearing release of the claim map, certificates, source systems, and environment lockfile is therefore an explicit submission gate rather than a completed claim of this draft.

Explicit non-results. This paper does not establish a global causal spacetime resolvent, a complete QNM expansion, a late-time asymptotic formula, a retarded inverse-Laplace deformation, a time-domain instability, finite Bondi flux for the generalized constant component, a standard asymptotically flat domain containing that component, excitation by a specified astrophysical source, detector sensitivity, an overtone tower, an all-multipole Bach nonsplitting theorem, or a quantum positivity/no-go theorem. It also does not establish the full coupled massive-system crosswalk or a physical massive-QNM slope. The polynomially weighted term is proved only for the isolated local resonance contour.

8 Conclusion

The repeated Regge–Wheeler factor in axial Schwarzschild pure Weyl gravity is not a direct sum. It is a non-split self-extension with the elementary projective class

$$[\mathcal{I}_{\mathrm{Bach}}] = \frac{i\omega}{2}[1 - 2/r].$$

That direction equals the scalar spin-two graded projection of the formal critical mass-squared jet. Its linearly growing rational primitive matches the differentiated massive phase, and the Coulomb exponent has vanishing first derivative. A full coupled-system and exact-Jost crosswalk remains an explicit gate.

At the certified damped Schwarzschild mode, the scalar QNM is simple but the extension selector is nonzero. The full connection is therefore defective. Its geometric root is Einstein; its generalized root has a nonzero Ricci-carrier quotient. The same intrinsic invariant controls the nonzero rank-one second-order pole of the exterior cut-off outgoing Green operator.

The leading pole survives an asymptotic observation test and a complexified source-space test. Its Einstein range has nonzero Bondi shear, and a radially compact conserved, traceless odd source can have strictly nonzero adjoint overlap. The enhanced isolated-contour profile is therefore an ordinary outgoing Einstein QNM multiplied by time, even though the constant generalized component has weakened asymptotic falloff.

This gives a Schwarzschild counterpart, in asymptotically flat spacetime, of the critical-gravity logarithmic-partner mechanism at the level of the filtered Bach resonance. The remaining work separates cleanly into two problems: the complete massive axial/Jost crosswalk, and a global causal domain supporting the differentiated endpoint state and a justified retarded contour deformation.

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